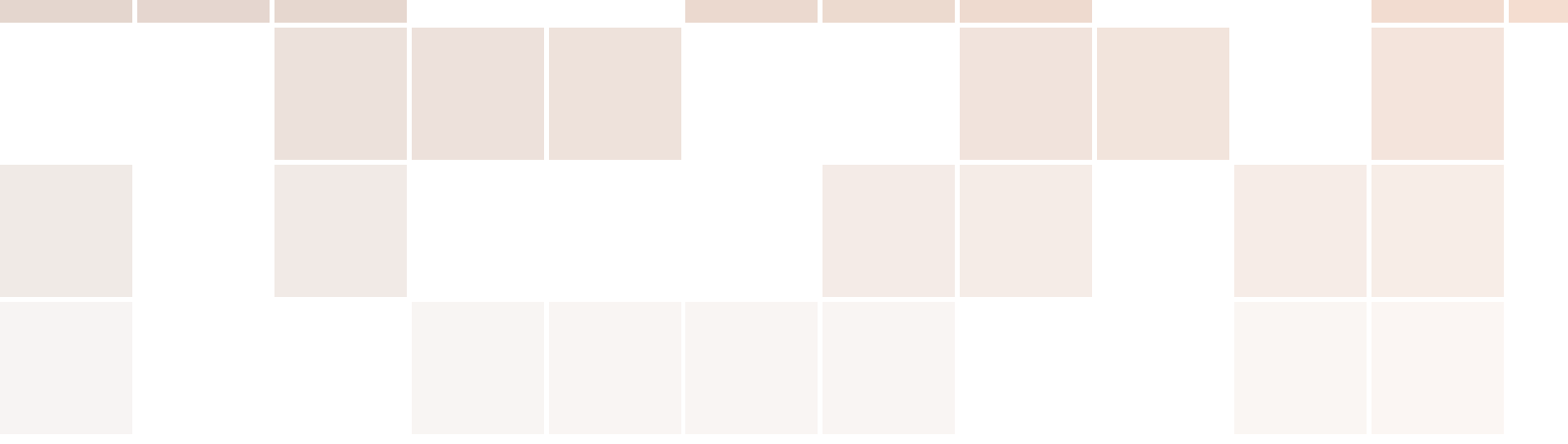
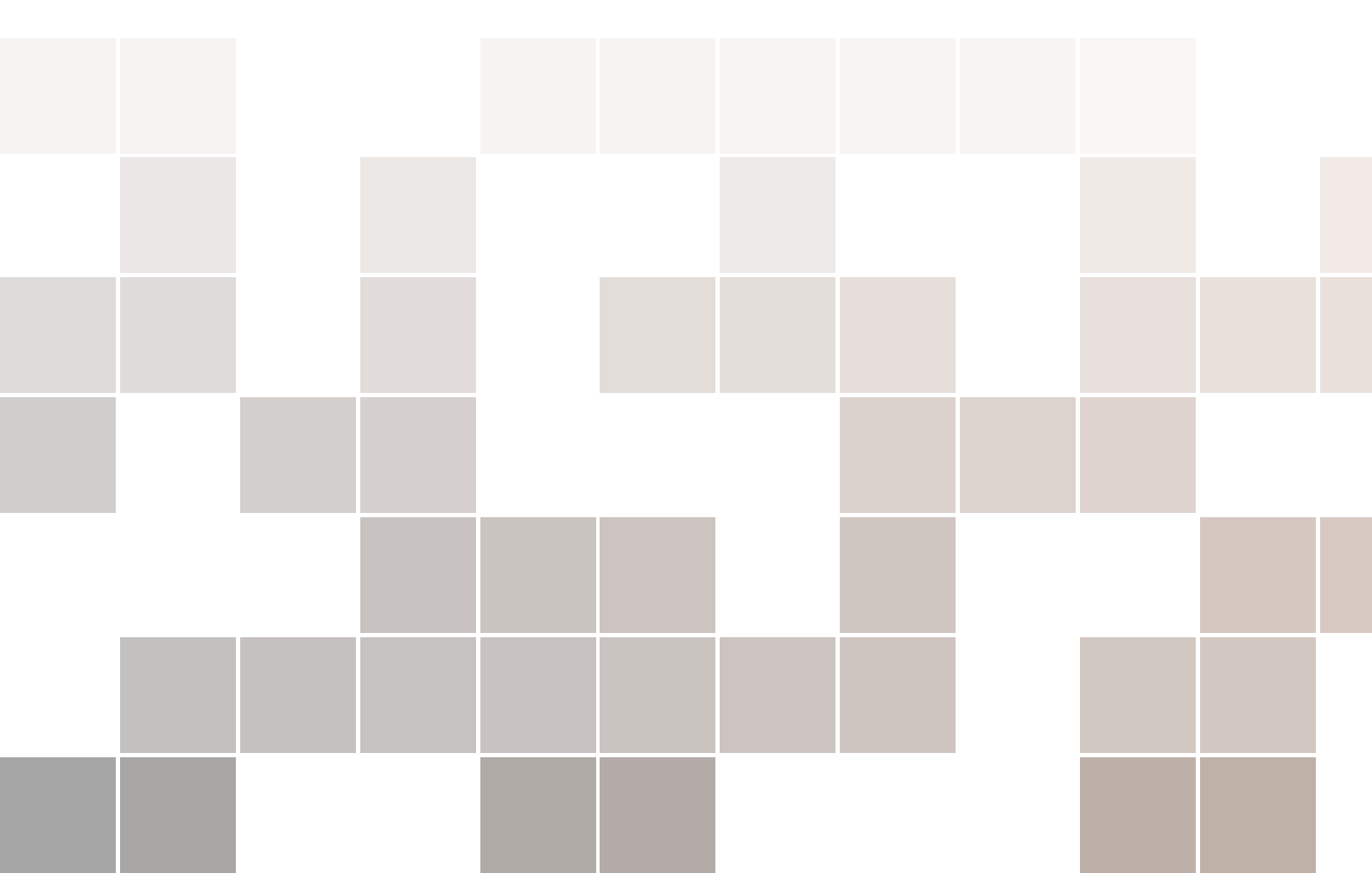

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ASTROPHYSICS COURSE PROJECT 2025 (PH-556)

Finding Azatoth

Ali Murtaza



Abstract

Supernovae (SNe), like Gamma ray Bursts (GRBs), are one of the most violent events that are happening in the cosmos so incredibly energetic that they outshine their host galaxy during the event and for a while after the event has happened. It entails the end of a star's story as it reaches its end after millions or billions of years of evolution in its catastrophic yet beautiful demise. Their luminosities typically reach $\sim 10^{43} \text{ ergs}^{-1}$, comparable to the typical luminosity of a galaxy in the optical wavelength, hitherto, outshining their entire host galaxy. SNe are one of the most important events in the universe as they are the origin of heavy elements found in the cosmos, they not only disperse heavy elements synthesized during hydrostatic stellar evolution into space but also create various elements like iron, cerium, lanthanides and especially gold in the famous r-nucleosynthesis, which also happens during GRBs. SNe are an amazing multi-messenger source as they produce signals in all different spectra, be it electromagnetic, Gravitational (GW) or neutrinos. They show diverse spectral properties which reveal the chemical composition of the ejected outer layers. There are different types of SNe based on their mechanism and spectral properties. Type Ia are a result of thermonuclear disruption of C+O white dwarf (WD). In contrast, the remaining classes arise from core collapse of massive stars, aptly named core collapse supernovae (CCSNe). Type II SNe originate from red supergiants, while Type Ib and Ic explosions correspond to progenitors that have lost their hydrogen rich cores. In case of type Ic which are also helium rich, envelopes stripped either by strong stellar winds or binary interactions, something I am also currently working on, which involved evolving stars and seeing what fraction of them turn into Neutron star- White Dwarf binaries (NS-WD), hence creating a physically acceptable population of NS-WD binaries within a good limit of the observed population currently, which sadly is 2 or 3 right now, but that's due to instrumentation limitations, with tools like LISA, DECIGO and BDECIGO we'll be able to measure the merger frequencies of these NS-WD systems and hence will get a better understanding of their population

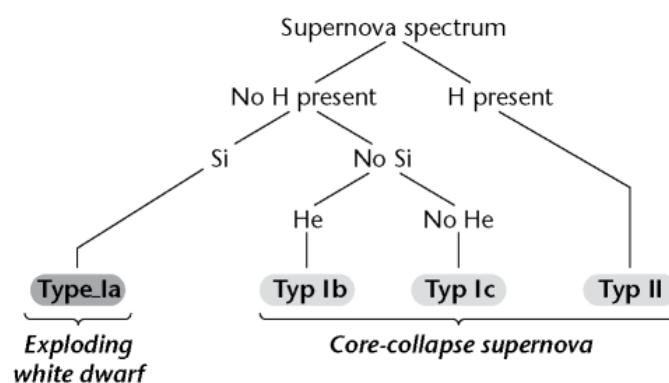


Figure 1: Classification of Supernovae based on their spectral properties. (https://ebrary.net/55392/sociology/supernovae_supernova_remnants)

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Part One

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1. Research Objectives and Target Selected

1.1 Research Objectives

The aim of the project is to study Type II supernovae. We will utilize early-time observations to constrain the progenitor's physical parameters like mass, radius etc. The analysis will be carried out using the data from the GROWTH-India Telescope (GIT). Here are our key objectives:-

Utilize early-time optical light curve peaks and color temperature measurements to directly estimate:

- Progenitor radius (R), which determines early shock breakout emission.
- Shock velocity (v_s), a critical parameter for explosion energy and luminosity scaling.
- Recover envelope and core mass.

1.2 Target Selected and observations

We plan to observe the recently discovered supernova SN 2025yyn, which was detected on 2025-09-30 at 11:27:30.997 UT by ZTF (1). The transient was reported at a magnitude of 18.31 in the "g" filter and classified as a Type II supernova by (2). Based on our visibility checks, the target is observable between 15:30:00 UT and 23:10:00 UT from GIT (see Figure 1.1). We propose to carry out our observations from 21:20:00 UT to 23:10:00 UT in the g, r, and i bands to monitor the evolution of the light curve. During this time window, the target remains above 30° altitude, satisfying the GROWTH-India Telescope observing criteria. Additionally, the moon-target separation exceeds 60°, which minimizes the risk of any significant lunar light contamination in the observations. Since our study focuses on the early peak behavior of Type II supernovae, we propose to conduct observations over a period of 20–24 days with a 2-day cadence to adequately capture this phase. As the supernova lies within its host galaxy Z333-5, we anticipate the need for image subtraction techniques to perform accurate photometry. Considering the time

constraints, we may not be able to do photometry with image subtraction. To eliminate the host background contribution, we will perform aperture photometry on the nearby background of the star and subtract it from the main source. However, for better modelling, we may request subtracted images to perform accurate photometry.

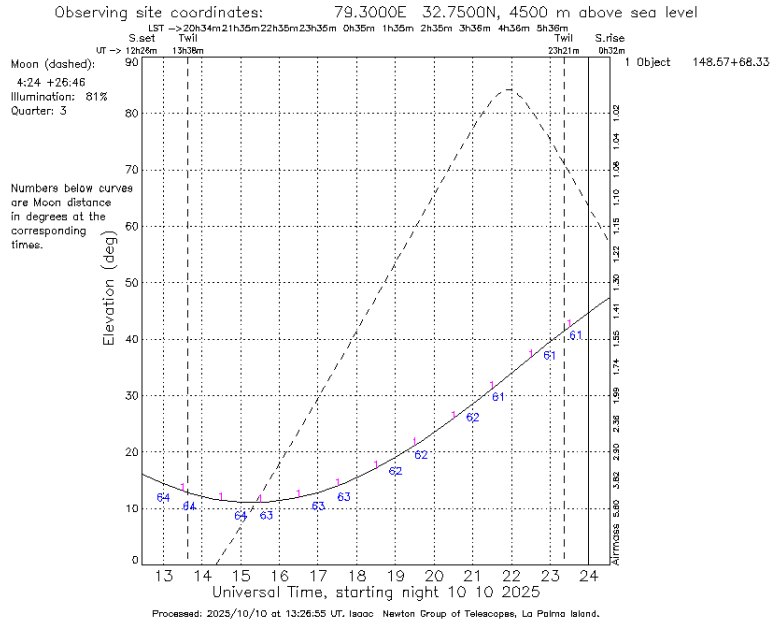


Figure 1.1: Visibility plot of the target SN 2025yyn as seen from the GROWTH-India Telescope.

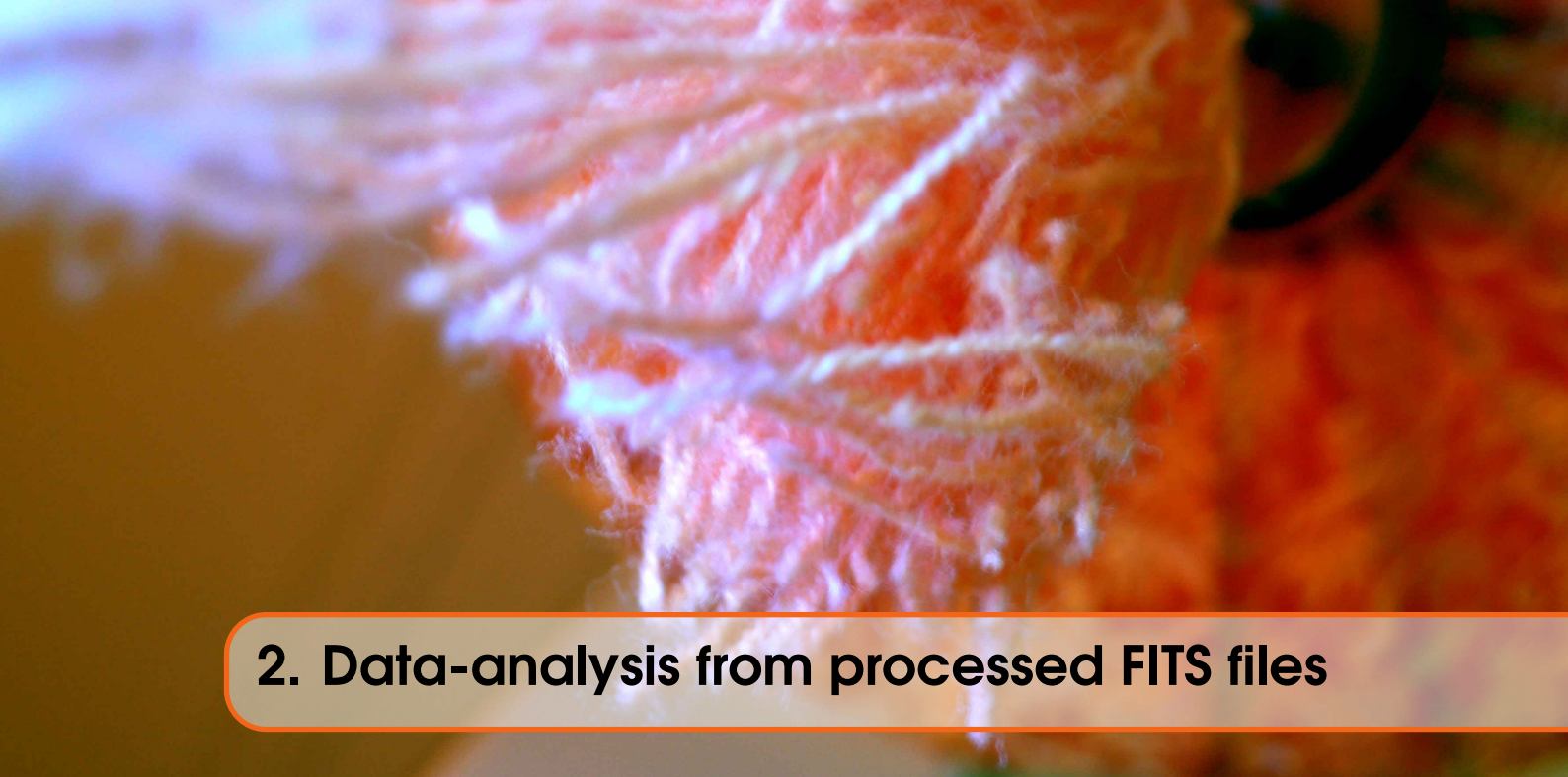
If sufficient observations to capture the early peak of SN 2025yyn are not available, we intend to shift our focus to SN 2023ixf. This supernova was discovered in the outer spiral arm of the host galaxy M101 on May 19, 2023, at 17:27:15.00 UT, with a magnitude of 14.9 in the “clear” filter (3). This supernova was then classified as a Type II (4). Several follow-up observations have been conducted, as it is one of the nearest core-collapse supernovae observed in the last 25 years. For this target also, we would need around 20 days of GIT observations in multiple bands with a cadence of 2 days to capture the early peak in the light curve. By analyzing this early peak using model, we can fit the light curve and extract valuable physical information about the progenitor star. These observations are crucial for constraining parameters such as the progenitor radius and shock velocity (5, 6). Early-time data significantly improve the precision of explosion energy estimates and other key physical properties.

We’ll be using **SN 2023 ixf** for our project as we were not able to get sufficient data for SN 2025 yyn considering the timeline of the project. All further analysis is done with data from GIT for **2023 ixf**

1.3 Software and Tools used

We will perform photometry on the GIT observations using softwares discussed in class, that is, **DS9** and **APT**, DS9 will help identify the source coordinates with catalog coordinates for conversion to image pixels and APT will be used to perform aperture photometry of the source and the catalog, the catalog, further discussion is done in the data-analysis section.

For the modeling, we will use publicly available Python package (https://github.com/griffin-h/lightcurve_fitting) that provide tools to fit analytical models to multiband light curves of astronomical transients. Specifically, we will implement the shock cooling model to fit the early-phase light curves of Type-II supernovae.



2. Data-analysis from processed FITS files

The **Sloan digital sky survey** (SDSS) is a major astronomical survey that has mapped millions of celestial objects using a dedicated 2.5-meter wide-angle optical telescope at Apache Point Observatory in New Mexico. It has revolutionized our understanding of the cosmos through its multi-phase data releases.

The **Flexible Image Transport System (FITS)** is the standard data format used in astronomy for storing and exchanging scientific data, including images and tables. Endorsed by NASA and the International Astronomical Union, FITS files consist of headers containing metadata and data units that store the actual scientific data. PHANGS utilizes FITS files to store and manage the extensive data collected from various telescopes, facilitating detailed analysis of galactic structures and star formation processes.

2.1 SNe 2023ixf

SNe 2023ixf was discovered in the outer spiral arm of the host galaxy M101 on May 19, 2023, at 17:27:15.00 UT, with a magnitude of 14.9 in the “clear” filter (3). This supernova was then classified as a Type II (4). Several follow-up observations have been conducted, as it is one of the nearest core-collapse supernovae observed in the last 25 years.

A type II SNe is classified by presence of Hydrogen lines in the spectrum of the SNe, we took several data points spanning 10-11 days to see the fluctuation in the magnitude in different bands namely, r, g and i and used these values to plot the lightcurve of SNe in different bands with error bars using aperture photometry using APT and DS9 and feed this into the publically available python package with (https://github.com/griffin-h/lightcurve_fitting) appropriate model to estimate the parameter of the progenator of the SNe

2.1.1 Data analysis for SNe 2023ixf

We had divided the work to calculate the lightcurve in different bands with r band done by Jyoti, g band being done by Abhinav and me doing the i band. Hitesh focused on understanding and using the model to fit these curves into in the meantime. The i band data for GIT was already processed in a FITS file, that can be used for our analysis.

The analysis follows the following steps:-

- Open .wcs.proc.fits file for a particular day in DS9, this will open the image in DS9 which we will use to identify ra,dec and pixel coordinates of the targets/object in question.
- Use a circular region to identify that region's coordinates, use the header option to get TARRA and TARDEC, this will give the target which is SNe 2023ixf ra and dec, in a popup table, make sure it is in degrees.
- Convert to image/pixel coordinate and now we're ready to use APT to perform photometry on the required target.
- Open the **same FITS file** in APT and check the settings for option B which has sky annulus-median subtraction, perform photometry and check the radial profile.
- The radial profile should take data before a sudden decay Sky-subtracted surface brightness (DN) , an example is shown below.
- Note the apparent magnitude for the target and the error associated with it.

Below is the radial profile for the target where the major radius of the annulus is 5, as we can see that the sky-subtracted surface brightness for major radius 5 has datapoints where the decay is not linear. We reduce the major radius of the annulus to 3 to get a better radial profile, which is shown below for the same file, this major radius is kept the same throughout for all further observations of the target. The same thing is done for calculations of the m_{inst} for the cataloged sources from SDSS 12, keeping the profile constant for all our sources.

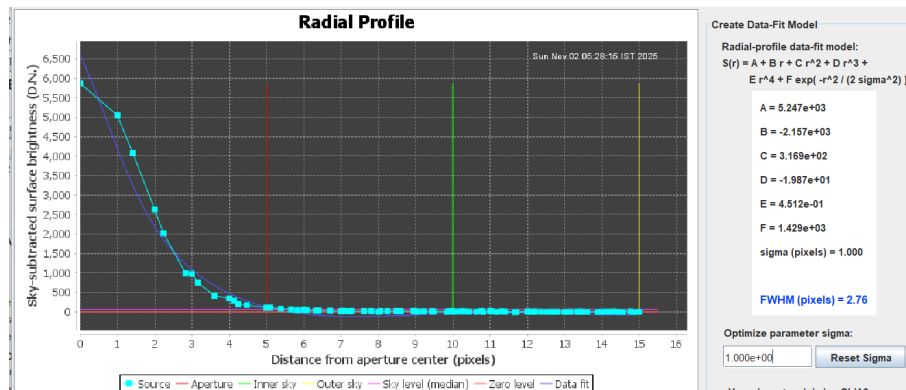


Figure 2.1: Radial profile of 20230520152114-977-RA.wcs.proc.fits with major radius 5.

Radial profile for target in the FITS file "20230520152114-977-RA.wcs.proc.fits" for major annular radius 3. This will give more accurate instrument magnitude as we are only performing photometry on the more localised source that is at the centre of the aperture. We only take values for a constant decay in brightness subtracting the transient parts, this gives a better value for magnitudes and lower error bars when done for zero point calculation using SDSS 12 Catalog.

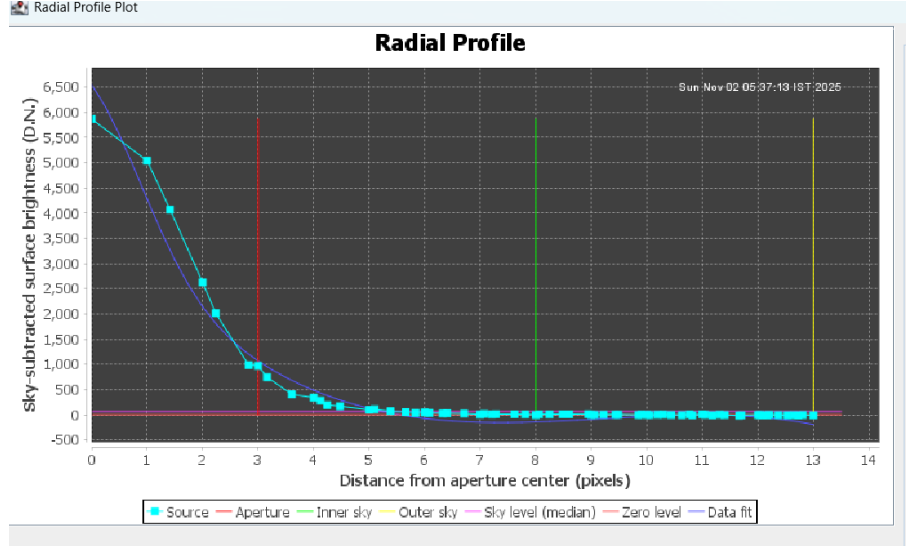


Figure 2.2: Radial profile of 20230520152114-977-RA.wcs.proc.fits with major radius 5.

2.1.2 Zero Point and it's calculation

In optical astronomy, the **zero point** is a calibration constant used to convert instrumental magnitudes (derived from raw detector counts) into standard magnitudes that align with a known photometric system. This ensures that measurements from different instruments and nights can be compared meaningfully.

The instrumental magnitude m_{inst} is calculated from the observed counts as:

$$m_{\text{inst}} = -2.5 \cdot \log_{10} \left(\frac{\text{counts}}{\text{exposure time}} \right)$$

This value is arbitrary and depends on the telescope, detector, filter, and observing conditions.

Zero Point Calculation

To calibrate the instrumental magnitude, we use a standard star with known magnitude m_{std} . The zero point ZP is calculated as:

$$ZP = m_{\text{std}} + 2.5 \cdot \log_{10} \left(\frac{\text{counts}_{\text{std}}}{\text{exposure time}} \right)$$

Calibrated Magnitude

Once the zero point is known, the calibrated magnitude m of any target object is given by:

$$m = m_{\text{inst}} + ZP$$

This was the... Labour that made me think of that famous line...

"TYCHO BRAHE DID THIS IN A CAVE WITH A BOX OF SCRAPS"

Not exactly as i remember it, but it made me truly appreciate him a lot more, spending over 10 years getting data from naked eye observation's in the night sky, with just coarse paper and charcoal. I say this because we just had to do it 10+ times, that too with the help of amazing tools like **google sheets**, **DS9** and **APT**, even using these it was incredibly tedious, we spent over 2 nights, entire nights just painstakingly calculating zero point of every catalogued source we chose.. over and over again. This made me appreciate their work from the bottom of my heart and i have newfound respect for my fellow astrophysicists.

It involves the following steps:-

- Use DS9 to open the same source file and through the analysis menu open the catalogues available and choose SDSS12,(which is something we didn't do before, we used GAIA, which i'll get back to in a few seconds).
- Now choose five sources which **You can see with your eyes** from the catalogue and note their Ra, dec, catalogue magnitude and error in catmag.
- Next for each individual source, we do the same thing we did for the target source, using DS9 paste ra and dec in the region, convert said region to image-pixel.
- Using the image coordinates use APT to perform photometry for each individual catalogued source while maintaining a good radial profile.
- **Don't change the annulus major radius from the first source** as changing it can cause huge magnitude variation for sources, keep it same for all sources and get the instrument magnitude and error.
- Do this for all the catalogued sources keeping the annulus major radius constant after first measurement/ reading.
- After getting the zero point for all five sources, get their median and add to the instrument magnitude of target, to get apparent magnitude for this **one** file/day. similarly, find the zero point error and get error in apparent magnitude
- Now, do the same steps for all other days, making sure you keep the same five sources for measurement of zero point, as the instrument magnitude for them will be different but their catalogued magnitude will be the same.

Doing this will give us the apparent magnitude for each day in i band which i was calculating. Plotting the magnitudes vs days with error bars gives me the lightcurve of the source in the i band which is what we will feed to the python pipeline to get to our goal, which is to find the mass and other physical properties of the progenitor.

2.1.3 Precautions

Some precautions to keep in mind while doing this process are:- (we learned the hard way)...

- Use SDSS over GAIA optical data as GAIA doesn't have the i filter, so the zero point we calculated was inconsistent since we used different catalog data (i used i from SDSS 12, and my team used GAIA for g and r filter), so we had to measure the zero points and calculate magnitudes using photometry again, 11+55 for everyone, 160+ times... I respect Tycho Brahe a lot more after this, and my dear observational astronomer as well.
- Make sure the radial profile is good enough for photometry.
- The sources chosen from the SDSS 12 catalogue should be visible to us decently bright, so when we do the photometry the source is clear in APT as well, otherwise we will get results that are way off from actual values.
- If the sources chosen from catalog are not well visible in a file, change the file to where they are visible so photometry can be done accurately.

2.1.4 Results from Photometry of different bands

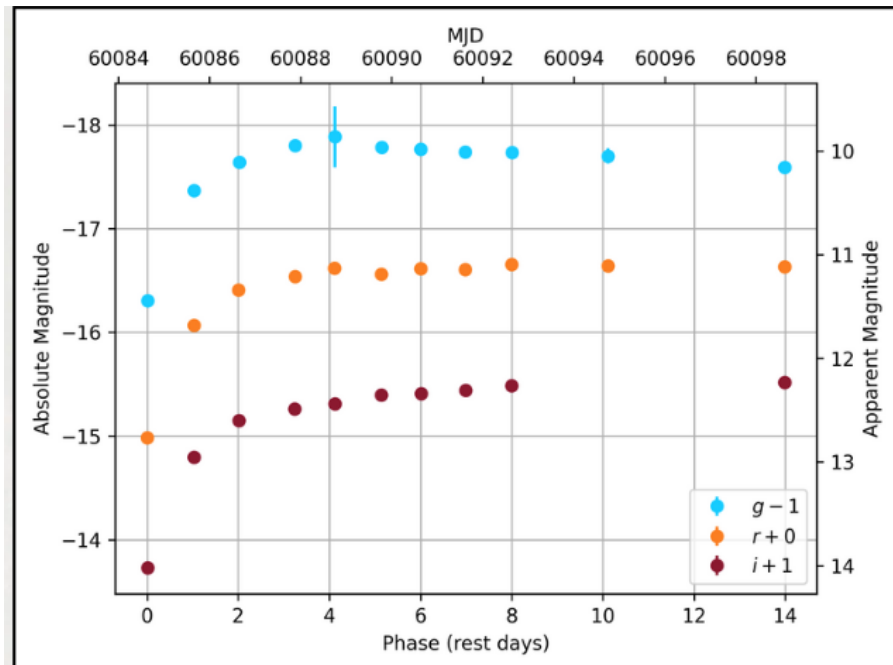


Figure 2.3: Lightcurve for r,g and i band scaled to see the different plots.



3. Markov chain - Monte Carlo (MCMC) using griff

Markov chain - Monte Carlo (MCMC) is a technique which is used to map probability distribution by sampling from a distribution of datapoints. In Bayesian inference, the distribution of interest is the "posterior probability", which represents our knowledge of the model's parameters after observing the data. The posterior is constructed from two components:-

- 1. The Log-prior (`prior.logp`): This function encodes our prior knowledge about the parameters before we see the data. The `light_curvefitting.model` module provides classes for common priors, such as `UniformPrior` (constraining a parameter between two hard bounds) and `LogUniformPrior` (a prior that is uniform in logarithmic space, often used for scale-invariant parameters like mass or radius. The log-prior function returns a finite value (e.g., 0) for allowed parameters and $-\infty$ (zero probability) for any parameter value outside the allowed range. This effectively blocks the unphysical regions of the parameter space, such as negative mass or radius or speeds exceeding or equal to the speed of light for shock velocity.
- 2. The Log-likelihood (`Model.log_likelihood`): This function quantifies how well the fit has been executed, It answers the question: "Given this specific set of parameters, what is the probability of observing our actual data?", typically this calculation assumes that the observational errors on data points are **Gaussian**. In this case, the log-likelihood is mathematically related to the standard chi-squared (χ^2) statistic ($\log \mathcal{L} \propto -\chi^2/2$).

3.0.1 Lightcurve_fitting using MCMC

We use the emcee sampler and this emcee sampler is fed to a single "log-probability" function, which, by Bayesian definition, is the sum of the two components defined above: $\log P(\text{posterior}) = \log P(\text{prior}) + \log P(\text{likelihood})$. The MCMC algorithm deploys "walkers" and these walkers then explore the multi-dimensional parameter space, preferentially spending time in regions with a high log-posterior

probability.

The Bayesian framework is encapsulated in high-level functions within the package, two of which are central to our fitting model namely:-

- `lightcurve_fitting.fitting.lightcurve_mcmc`: This is the function used for phase 1 analysis to produce the results. It is designed to fit a complex, analytical model to entire light curve. Its inputs include the LC (light curve), a Model object (e.g., ShockCooling), and the MCMC run parameters, such as the number of walkers (nwalkers), burn-in steps (nsteps_burnin), and production steps (n_steps)
- `lightcurve_fitting.bolometric.calculate_bolometric`: This function is used for the Phase 2 analysis. It internally calls a MCMC sampler (spectrum_mcmc) if the `do_mcmc=True` flag is set. The function first groups the light curve data by epoch (using `group_by_epoch`) and then fits a blackbody model to each epoch's SED to find the optimal Temperature (T) and Radius (R)

3.0.2 Results from MCMC routine

We used Markov chain - Monte carlo (MCMC) approach to fit the data into the blackbody plot we get from Planck's distribution:-

$$B(\nu, T) = \frac{2h\nu^3}{c^2} \cdot \frac{1}{e^{\frac{h\nu}{k_B T}} - 1}$$

Wikipedia. This model fitted the lightcurve we got into this equation for different data points to get the normalization factor, from which we get the temperature of the "ejecta mass" and shock velocity using the formulae:-

$$\rho_0 = \frac{3f_p M}{4\pi R^2} \delta^n$$

(5) where

f_p is a numerical factor of order unity

M is the ejecta mass (any remnant is neglected)

R is the stellar radius

$\delta = \frac{R-r}{R}$ (fractional depth from the stellar surface)

$n = \frac{3}{2}$ (polytropic index for convective envelopes)

and finally using the formulae for shock velocity:-

$$v_{sh} = v_s^* \delta^{-\beta_n}$$

(5) where,

v_s^* is a free parameter

$\beta_n = 0.191$ (a constant)

Parameters used for MCMC routine

Table 3.1: Model Parameters and Prior Ranges

Variable	Prior Type	Range (Min, Max)	Initial Guess	Units
v_{ss}	Uniform	(0.0, 10.0)	(2.1–2.6)	$10^{8.5} \text{ cm s}^{-1}$
M_{env}	Uniform	(0.0, 5.2)	(2.8–3.0)	$M_{\odot}$
$f_p M$	Uniform	(0.0, 10.0)	(0.1–4.0)	$M_{\odot}$
R	Uniform	(0.0, 2.0)	(0.2–0.6)	10^{13} cm
t_0	Uniform	(60081.9, 60085.98)	(60083.9–60084.93)	MJD

We use the following burnin, steps and walkers for the MCMC routine

Table 3.2: Run Parameters

Parameter	Value
nburn_in	100
n_steps	6000
Walker	30

3.1 Results and Plots

The plot we got using the MCMC is

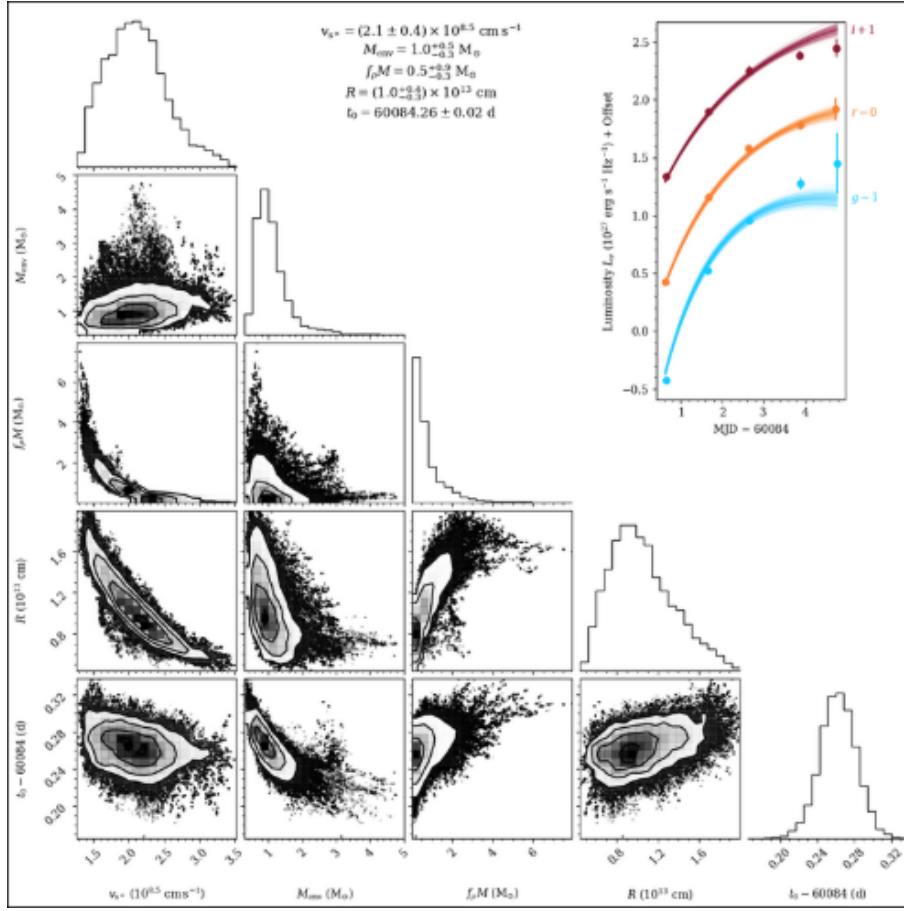


Figure 3.1: Posterior probability distributions for the five free parameters of the model, derived from an MCMC analysis. The diagonal panels show the 1D marginalized posterior histogram for each parameter. The off-diagonal panels show the 2D joint posterior distributions, with contours indicating the 1, 2, and 3-sigma confidence levels. The inset plot (top right) shows the best-fit model light curves (solid lines) with 1-sigma uncertainties (shaded regions) compared to the observed data points in the g, r, and i bands (offset for clarity). The text inset lists the median values and 1-sigma uncertainties for each parameter.

- Histograms on the diagonal represent 1D histograms, these show marginalized posterior probability distribution for a single parameter. It's the final value we have for one particular parameter, showing the full range of plausible values and where most probable values will be. The peak of the histogram is the most likely value.
- 2D plots show the posterior distribution for pairs of parameters. It means they reveal the correlation with the parameters, if the shape is circular the parameters are independent, if it's tilted or elongated, the parameters are correlated. For example, the plot for envelope mass vs shock velocity, the contour is tilted showing an anti-correlation
- The light curve fit shows the best fit model for the light curves generated using the most probable parameters. The shaded regions represent the 1-sigma

uncertainty in the model, it's generated by taking many parameter sets from high-probability regions of the corner plot and plotting all the resulting light curves, showing range of good fits.

3.1.1 Resulting parameters and comparison with studies

Table 3.3: Best-fit physical parameters of SN 2023ixf. MSW23 stands for the model parameters derived by Morag et al 2023.(5)

Parameter	Finding Azatoth	MSW23	Units
Shock velocity (v_{s*})	6.64 ± 1.3	$7.2^{+0.5}_{-0.4}$	10^3 km/s
Envelope mass (M_{env})	$1.0^{+0.5}_{-0.3}$	5^{+2}_{-1}	$M_{\odot}$
Ejecta mass $\times$ numerical factor ($f_p M$)	$0.5^{+0.9}_{-0.3}$	$1.6^{+0.7}_{-0.5}$	$M_{\odot}$
Progenitor radius (R)	144^{+57}_{-43}	410 ± 10	$R_{\odot}$
Explosion time (t_0)	84.26 ± 0.02	$83.983^{+0.002}_{-0.003}$	MJD - 60,000

We can infer from the values we have gotten vs the value they have gotten that our values match for some parameters like, shock velocity and explosion time but for parameters like envelope mass and progenitor radius (R) we are getting very big deviations, this is because we have taken measurements using GIT only in the r, g and i band, where as they have taken un multiple bands like z,v,uv etc. and taken multiple measurements in the same day giving them the best data for fitting. The model we use can only take the first 5 days data and fit it, and our data was spread out as, 1 data point each day while they had more measurements in one day, this could be a possible reason for this mismatch. Another important thing is they also took data in the higher frequency bands like the violet, ultra -violet (uv) band, and since the explosion initially will produce maximum energy output i.e brightness or luminosity in the higher frequency bands like us band, therefore, using the uv data in their model could be a reason for them getting much better values then us for some parameters.

3.1.2 Luminosity and Color

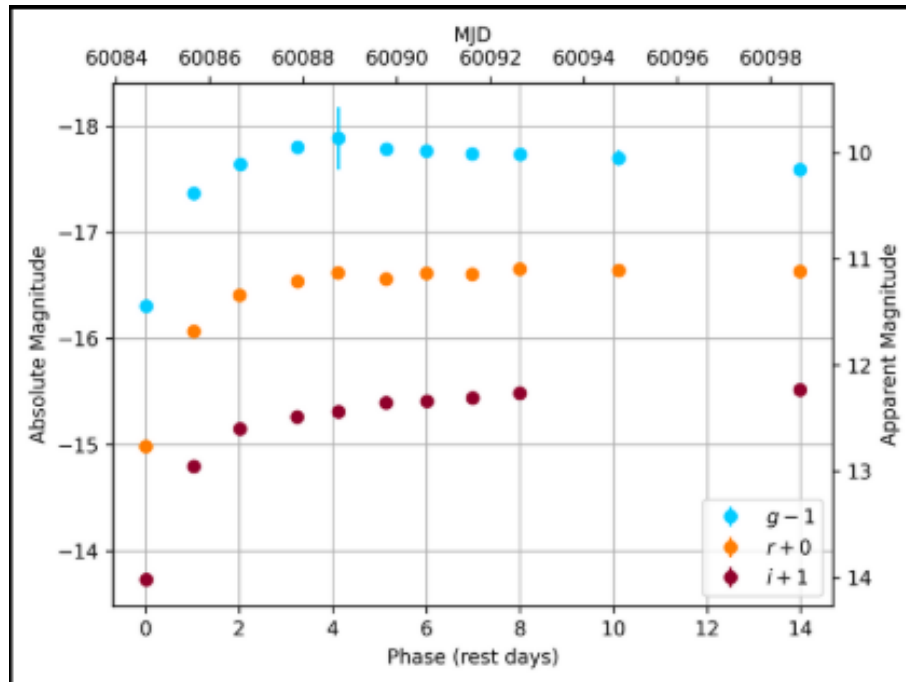


Figure 3.2: Lightcurve for r,g and i band scaled to see the entire lightcurve in all bands

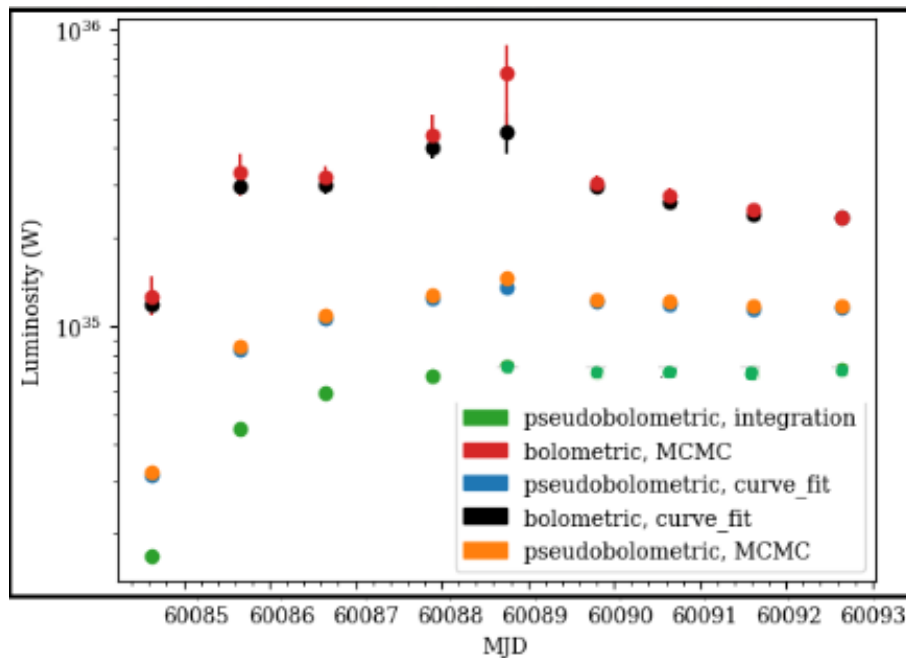


Figure 3.3: Luminosity (W) vs. Time (MJD), comparing bolometric (red, black) and pseudobolometric (green, blue, orange) estimates derived from different computational methods (integration, MCMC, and curve_fit)

Pseudobolometric luminosity refers to the luminosity that we did only in the bands

data we had, i.e r,g and i while bolometric luminosity refers to the luminosity of the entire spectrum.

3.1.3 Evolution of Temperature and radius of the shock

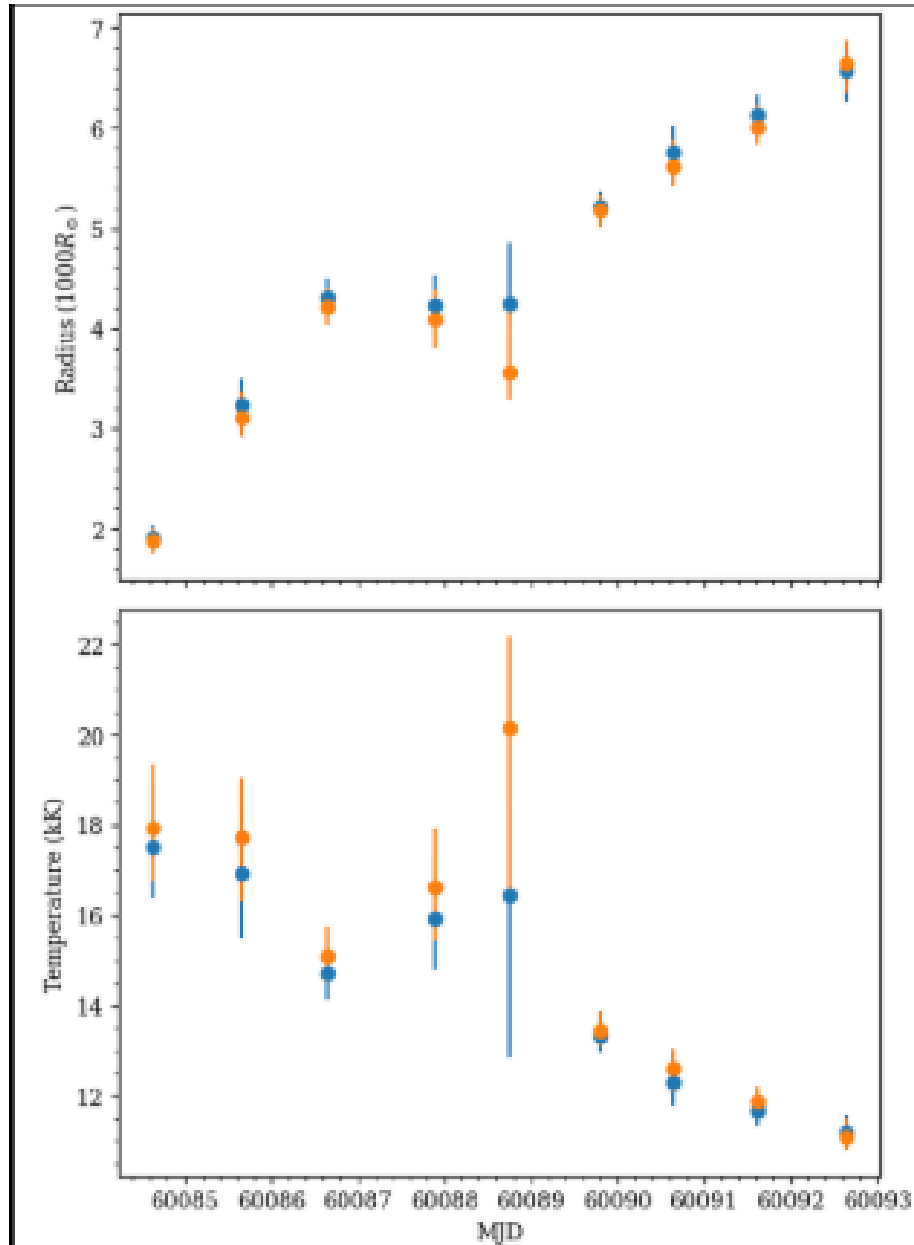


Figure 3.4: Time evolution of radius (top) and temperature (bottom) for the expanding shock from the SNe, plotted against Modified Julian Date (MJD). The top panel shows radius in units of $1000R_{\odot}$ and the bottom shows temperature in kilo-Kelvin (kilo_Kelvin)

This plot confirms our hypothesis that the shock is "cooling" while expanding as the temperature of the shock is seen having a downwards trend while the radius of the shock is seen to have an upward trend.

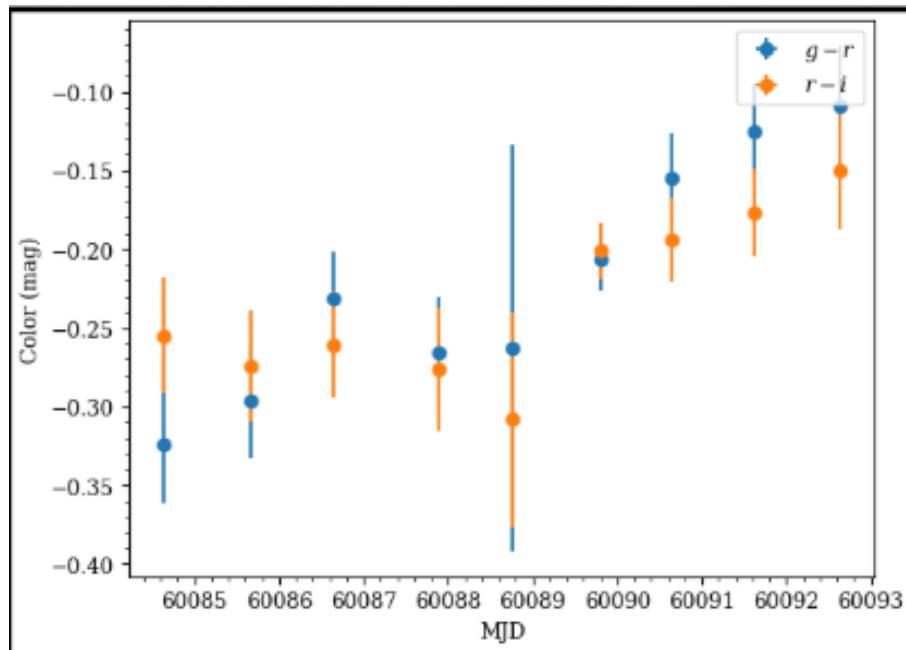


Figure 3.5: Color light curves for the $g-r$ (blue diamonds) and $r-i$ (orange circles) indices. The colors, measured in magnitudes, are plotted with their 1-sigma error bars against the Modified Julian Date (MJD). The plot illustrates a clear reddening trend, with both colors evolving from approximately -0.3 mag to -0.15 mag over the 8-day period

This plot also confirms our hypothesis that the shock is cooling, hence the "reddening" of the data.

3.2 Conclusion and future scope

Table 3.4: Conclusion

Topic	Key Findings & Results
Multi-band Photometry	Light curves were generated in the g, r, and i bands using Growth India Telescope (GIT) observations.
Best Fit Parameters	The light curve modeling yielded: • Shock velocity: 6640 km/s • fpM: $0.5 M_{\odot}$ These two parameters are consistent with morag et al 2023. (5). • Progenitor radius: $144 R_{\odot}$ • Envelope mass: $1 M_{\odot}$ These two parameters are inconsistent with Hosseinzadeh et al. 2023 (7)
Temporal Evolution	The analysis confirmed the object's physical evolution: • Radius: Expanded significantly, from $\sim 2000 R_{\odot} \rightarrow 7000 R_{\odot}$. • Temperature: Declined from $\sim 18 \text{ kK} \rightarrow 10 \text{ kK}$. • Color: Showed an increasing (reddening) trend, which independently confirms the temperature decrease.
Future Scope	To improve the accuracy of the model fitting, improved data sampling and additional observations especially in the UV and Violet band are required.



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